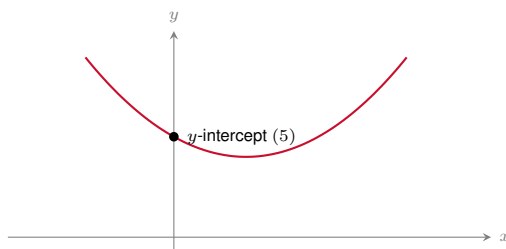


The numbers nobody wanted, and the words of the week

THE ONE IDEA

Once you allow the numbers nobody wanted, every polynomial has exactly as many roots as its degree. The roots may be imaginary; the answers you want are real.

THE PICTURE: A QUADRATIC THAT NEVER REACHES THE AXIS



$$y = x^2 - 2x + 5, \Delta = -16: \text{ no real roots.}$$

Roots $1 \pm 2i$: a conjugate pair, mirror images.

Vieta still works: sum 2, product $(1 + 2i)(1 - 2i) = 5$.

Icelandic: *tvíntala*, twin-number. A pair.

THREE SENTENCES, THREE STORIES

Sentence	Why it is true	Who paid for it	What it does this week
Always roots	$\deg f = n \Rightarrow n \text{ roots}$	Gauss, 1799: with complex numbers allowed, every polynomial factors completely into n linear factors.	The factor theorem and Vieta work because of <i>always</i> . Example 6's cubic has one real root and two complex ones.
Imaginary roots, real answers	they cancel	Cardano, 1545, met $\sqrt{-121}$ inside the cubic formula and flinched. Bombelli, 1572, carried it through: $(2 + i)^3 = 2 + 11i$, and the two cube roots add to 4, the real root.	Descartes, 1637, named them imaginary as a sneer, in the book that gave every point an address. Gauss, 1831, renamed them complex.
Imaginary roots wobble	$u_{n+1} = u_n - u_{n-1}$	The sequence below is a wave made of whole numbers; its characteristic quadratic $r^2 - r + 1 = 0$ has roots $\frac{1 \pm i\sqrt{3}}{2}$, of size 1, at 60 degrees, and 60 goes into a turn six times.	Pendulums, alternating current, business cycles: imaginary roots in the equation, real oscillation in the world.

THE DEMONSTRATION (DO IT BY HAND, ONE MINUTE)

Each term is the previous term minus the one before: $u_0 = 1, u_1 = 1$, then $1 - 1 = 0, 0 - 1 = -1, -1 - 0 = -1, -1 - (-1) = 0, 0 - (-1) = 1, 1 - 0 = 1$. The sequence is 1, 1, 0, -1, -1, 0, 1, 1, 0, ... forever, period 6. Nothing but subtraction, and it oscillates. The roots knew the period before any term was computed.

THE WORDS OF THE WEEK, IN ENGLISH AND ICELANDIC

parabola: placed beside, the throw-arc, *fleygbogi*. vertex: the turn, top point, *toppunktur*. discriminant: the teller-apart, *aðgreinir*. polynomial: many terms, *margliða*. degree: the highest power, *stig*. coefficient: the passenger on the variable, *stuðull*. root: zero-place, *núllstöð*. factor: *þáttur*. remainder: leftovers, *afgangur*. imaginary: Descartes's sneer; complex number: *tvíntala*, twin-number.

IN PLAIN ENGLISH

When the curve never reaches the axis, the roots did not vanish. They moved somewhere the real line cannot see, in a mirror pair, and everything that depends on them, the factoring, the sum, the product, goes on working.

TRAPS

- $\sqrt{-16} = 4i$, not -4 . Then $x = \frac{2 \pm 4i}{2} = 1 \pm 2i$: divide both parts by 2.
- Conjugate pairs need real coefficients. $x^2 - 2ix - 1 = 0$ has the repeated root i , no pair.
- The product of a conjugate pair is $a^2 + b^2$, always positive: $(1 + 2i)(1 - 2i) = 1 + 4$, not $1 - 4$.

REMEMBERING: WHAT THIS WEEK LEANED ON FROM TOPIC 1

- 1.2 Sequences: the demonstration is a sequence defined by a rule; the characteristic equation is the quadratic that governs it.
- 1.11 and 1.12: $i^2 = -1$, conjugates, and the conjugate-root fact from Exam I, now with its reason: real sum, real product.
- 1.9 Induction: the sequence repeats every six steps; proving that for all n is an induction in the teacher's three-step layout.

Questions to ask in class

Ask at least one. The first is safe, the second shows you followed, the third is the one you actually want answered. Write down what was said; it is the part you will remember.

1. *[safe]*

In Example 6 the cubic had one real root and a quadratic factor with $\Delta < 0$. What are the other two roots, and are they a conjugate pair?

What you expect: $\frac{1 \pm i\sqrt{3}}{2}$, a conjugate pair. Gets the link between 2.11 and 1.12 said in class.

What he said:

2. *[shows you followed]*

Vieta gives the sum and product of the roots from the coefficients. If the coefficients are real, does that force complex roots to come in conjugate pairs?

What you expect: Yes: a real sum and a real product can only come from a mirror pair. He may answer with the quadratic formula instead; both are right.

What he said:

3. *[the one you want answered]*

Why were imaginary numbers invented, if a quadratic with no real roots is a perfectly good answer? Was there a problem that could not be finished without them?

What you expect: The cubic, not the quadratic: Cardano and Bombelli, 1545 to 1572, needed them to reach real roots. If he does not know, that is fine; the episode told it.

What he said:

Later. Polar form (1.13 to 1.15, later in the year): where multiplying two complex numbers becomes stretch-and-turn, and why these numbers describe anything that rotates or oscillates. Saved for then.

Next. Next week: 2.3 Functions, the Icelandic fall, the stone from the tower. HL on 14 October: Vieta again. Sprint 3 is due Friday 20:00; Quiz 1 on Saturday before the Zoom.